

Sickle Cell Disease (SCD) Screening and Education Initiative

Jaramogi Oginga Odinga University of Science and Technology (JOOUST)

Executive Summary

In May 2026, Jaramogi Oginga Odinga University of Science and Technology (JOOUST) together with partners conducted a **Sickle Cell Disease (SCD) Screening and Education Campaign**, reaching **242 participants**. The initiative combined health education with Hemoglobin screening to increase awareness of SCD, identify individuals with abnormal Hemoglobin variants, and promote early counseling and linkage to care.

The findings demonstrate that the campaign successfully engaged a predominantly student population while identifying individuals with sickle cell trait and sickle cell disease who would benefit from additional counseling, confirmatory testing, and ongoing clinical follow-up.

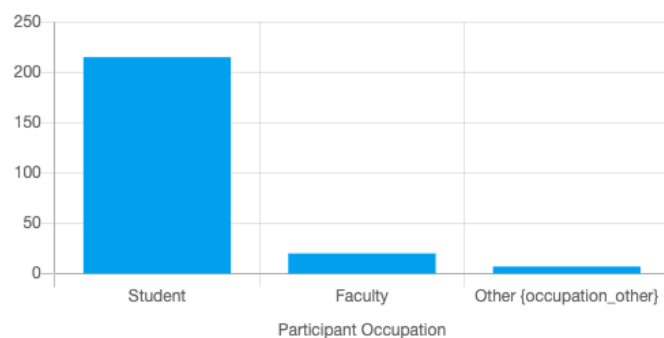
Key Highlights

Participation

- **242 participants** were screened and educated.
- The majority of participants were **young adults (18–25 years)**, reflecting the university student population.
- Most attendees were students, with smaller numbers of faculty participants.

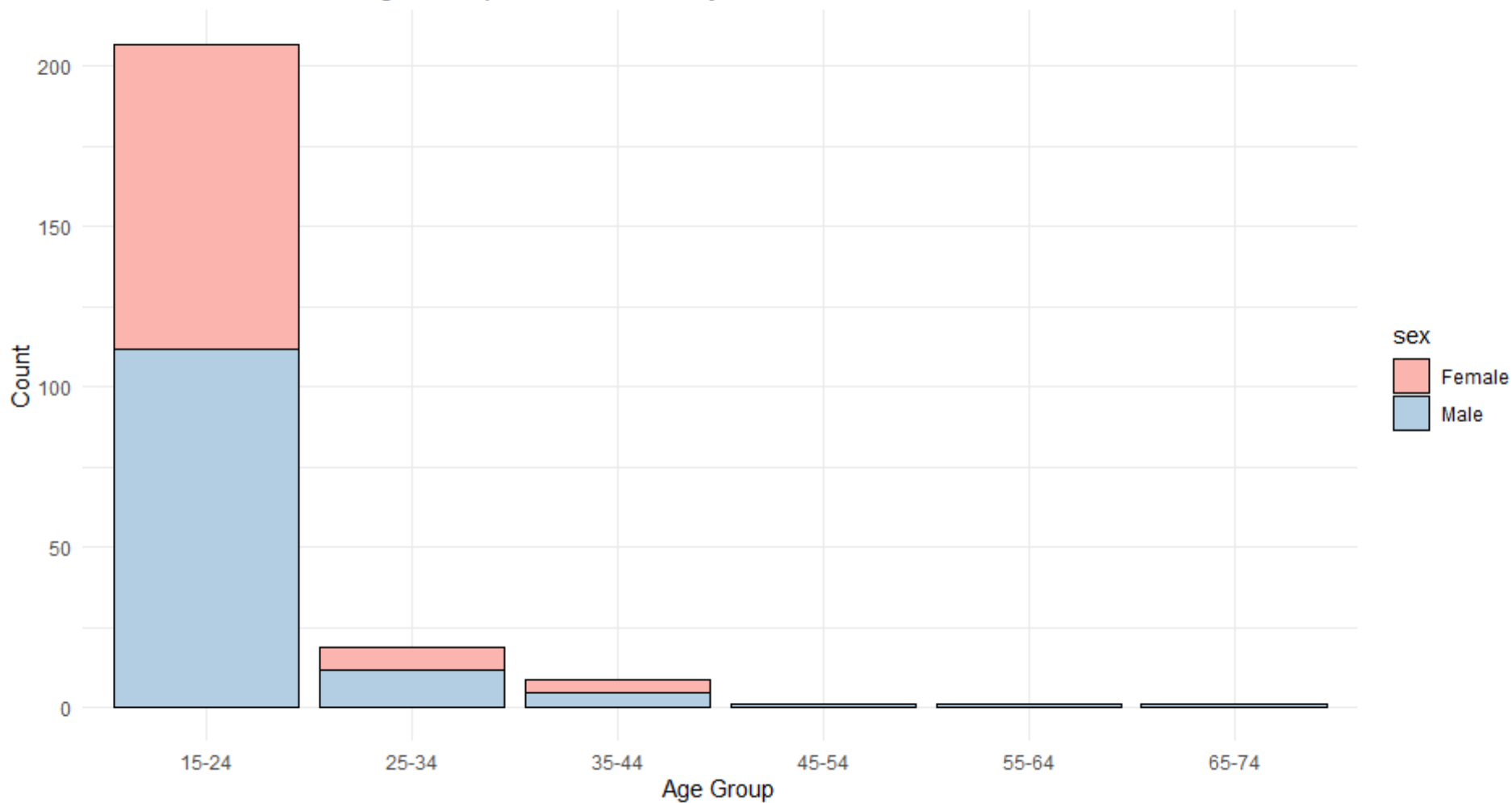
DEMOGRAPHICS

Occupation



Occupation results match the mean/median age results, mostly college students.

Stacked Bar Chart of Age Groups of attendees by Sex

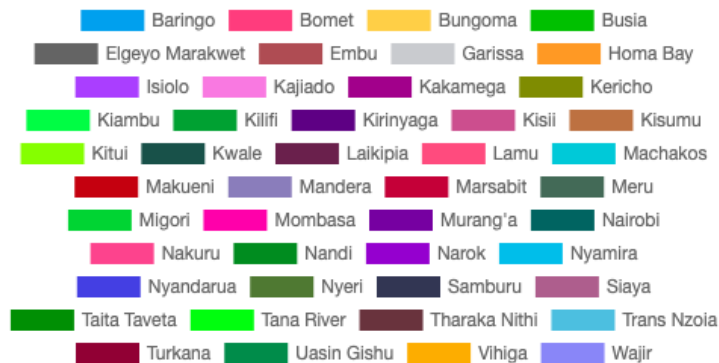


Geographic Reach

Participants represented multiple counties across Kenya, demonstrating that JOOUST attracts learners from diverse regions while maintaining a strong representation from **Siaya County**. This positions the University as an important platform for expanding SCD awareness regionally and nationally.

County of Residence

(click on the graphic to expand)

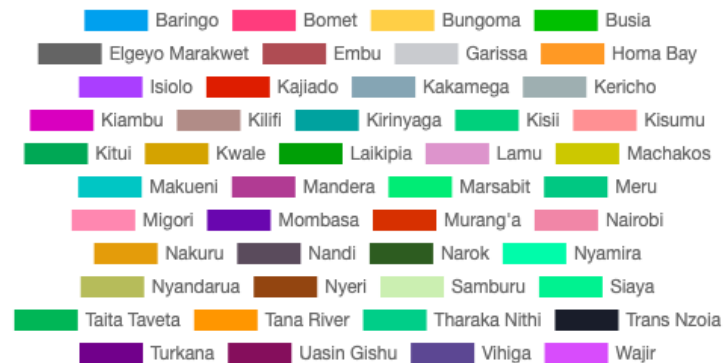


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Nearly **50%** of the group resides in **Siaya county**.

County of Birth

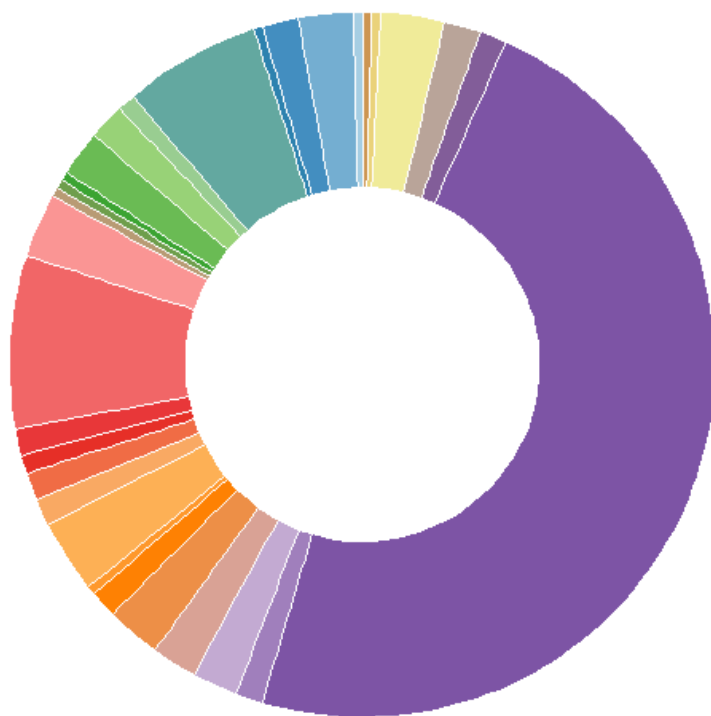
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The variation in county of birth (relative frequencies are no higher than 15%, contrast that with relative frequencies for county of residence) matches with the fact that this is a targeted college-aged audience most likely attending from a single university.

County of Residence Donut Chart with Percent Table

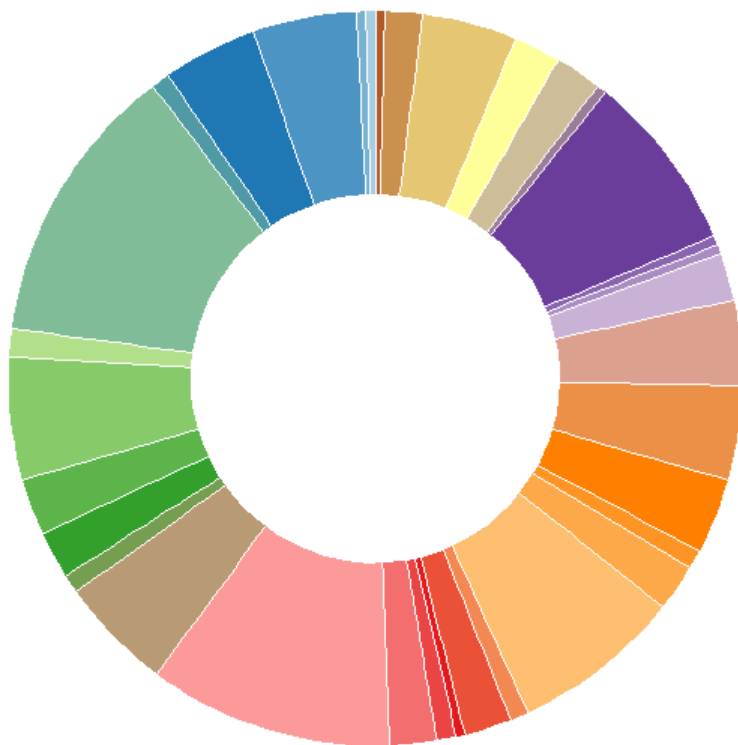


county_of_residence

Bomet	Makueni
Bungoma	Meru
Busia	Migori
Elgeyo Marakwet	Mombasa
Homa Bay	Murang'a
Kajiado	Nairobi
Kakamega	Nakuru
Kericho	Nandi
Kiambu	Nyamira
Kilifi	Siaya
Kirinyaga	Trans Nzoia
Kisii	Turkana
Kisumu	Uasin Gishu
Kwale	Vihiga
Laikipia	West Pokot

county_of_residence	percentage
Siaya	47.9
Kisumu	7.9
Homa Bay	6.2
Migori	3.3
Uasin Gishu	2.9
Kisii	2.9
Nairobi	2.5
Bungoma	2.5
Nandi	2.1
Kericho	2.1
Nakuru	2.1
Kakamega	1.7
Busia	1.7
Turkana	1.7
Murang'a	1.2
Trans Nzoia	1.2
Nyamira	1.2
Kwale	1.2
Meru	1.2
Makueni	1.2
Kajiado	0.8
Laikipia	0.8
Kilifi	0.4
Mombasa	0.4
Kiambu	0.4
Elgeyo Marakwet	0.4
West Pokot	0.4
Vihiga	0.4
Bomet	0.4
Kirinyaga	0.4

County of Birth Donut Chart with Percent Table



county_of_birth

Baringo	Meru
Bomet	Migori
Bungoma	Mombasa
Busia	Murang'a
Elgeyo Marakwet	Nairobi
Homa Bay	Nakuru
Kajiado	Nandi
Kakamega	Nyamira
Kericho	Nyandarua
Kiambu	Nyeri
Kilifi	Siaya
Kisii	Tana River
Kisumu	Trans Nzoia
Kwale	Turkana
Laikipia	Uasin Gishu
Machakos	Vihiga
Makueni	West Pokot

county_of_birth	percentage
Homa Bay	12.4
Kisumu	10.8
Siaya	7.9
Migori	7.5
Kakamega	5.4
Kisii	5.0
Bungoma	4.6
Uasin Gishu	4.1
Nakuru	4.1
Busia	4.1
Nandi	3.7
Nairobi	3.3
Kericho	2.5
Nyamira	2.1
Mombasa	2.1
Kwale	2.1
Trans Nzoia	2.1
Makueni	2.1
Turkana	2.1
Kiambu	2.1
Vihiga	1.7
Kajiado	1.2
Murang'a	0.8
Meru	0.8
Laikipia	0.8
Kilifi	0.8
Elgeyo Marakwet	0.8
Nyandarua	0.4
Tana River	0.4
West Pokot	0.4
Machakos	0.4
Baringo	0.4
Bomet	0.4
Nyeri	0.4

Health Insurance

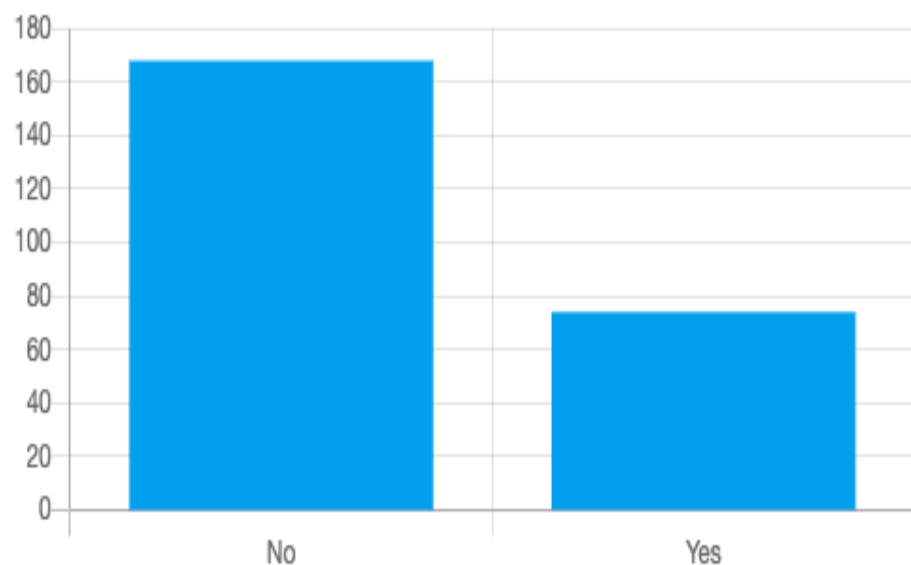
The screening identified a significant gap in health insurance coverage:

- Approximately **one-third (about 31%)** of participants reported having active health insurance.
- Approximately **69%** lacked health insurance coverage.

This finding highlights potential barriers to accessing regular care and long-term management for students diagnosed with sickle cell conditions.

INSURANCE

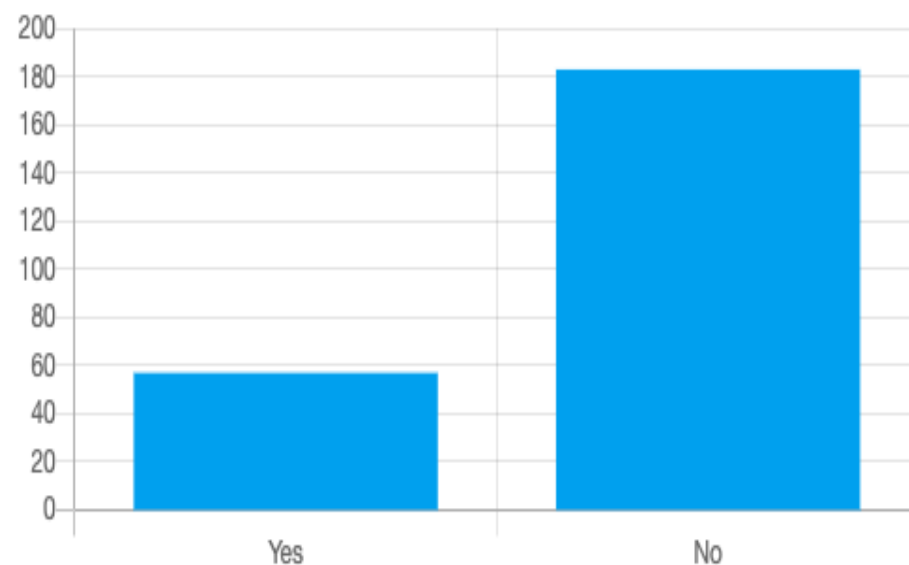
Covered Status



Are you under a...I cover?

About 84% of those with insurance coverage had SHA/SHIF (Social Health Insurance Fund), only 31% of the entire group had coverage.

Active Status



Is the cover cu...active?

About 24% of attendees had **active coverage**.

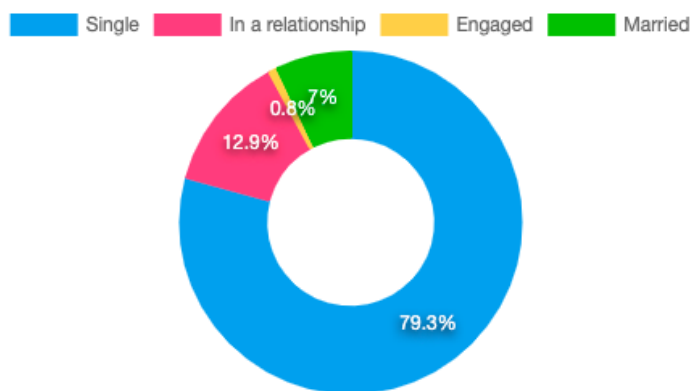
Knowledge and Awareness

The campaign revealed several important knowledge gaps:

- Many participants **did not know their partner's genotype**, limiting informed reproductive decision-making.
- Awareness of relatives living with sickle cell disease varied, suggesting opportunities to strengthen family-centered education and cascade screening.

RELATIONSHIP STATUS

Most attendees are single

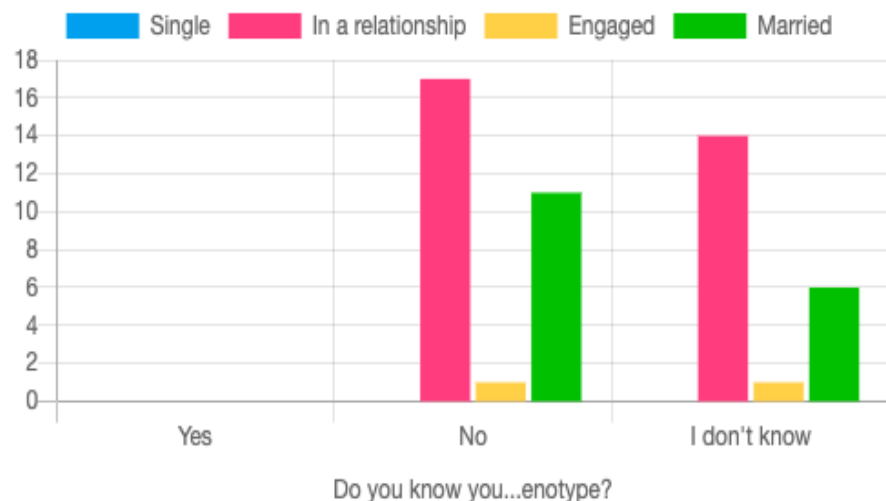


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About 4 out of 5 attendees reported being single.

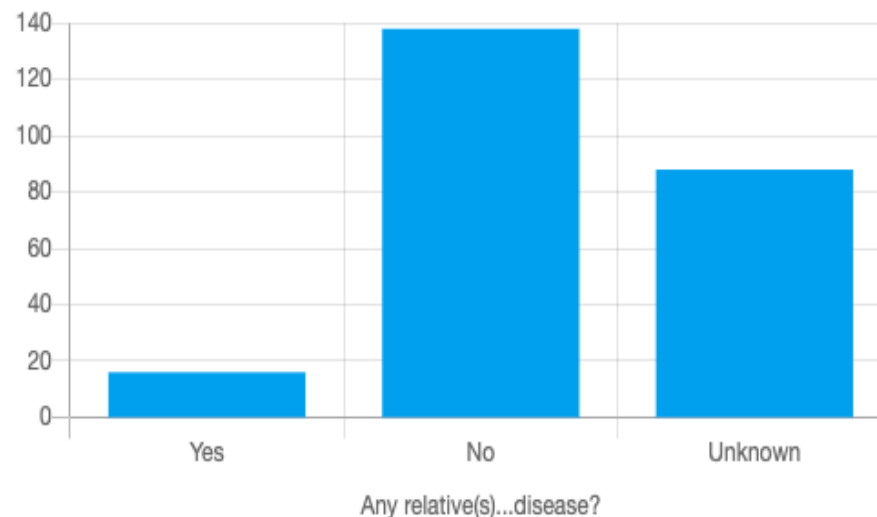
SCD KNOWLEDGE

Knowledge of Partner's Genotype



No one reported knowing their partner's genotype regardless of the stage of their relationship.

Knowledge of a Relative with a SCD Diagnosis



More than 1 in 3 (36% of) attendees did not know, and 7% knew of a relative with a SCD diagnosis.

Relationship of the Relative

	Count
Cousin	6
Aunt or Uncle	5
Sibling	5

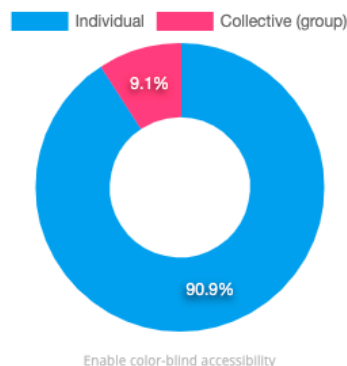
Counseling Preferences

Participants demonstrated overwhelming support for personalized counseling:

- Approximately **91% preferred individual counseling** over group counseling. This finding supports future investment in confidential, one-on-one genetic counseling services within university health programs.

COUNSELLING PREFERENCE

Counselling



Around **91%** of the cohort **prefers individual counselling**.

Hemoglobin Screening Results

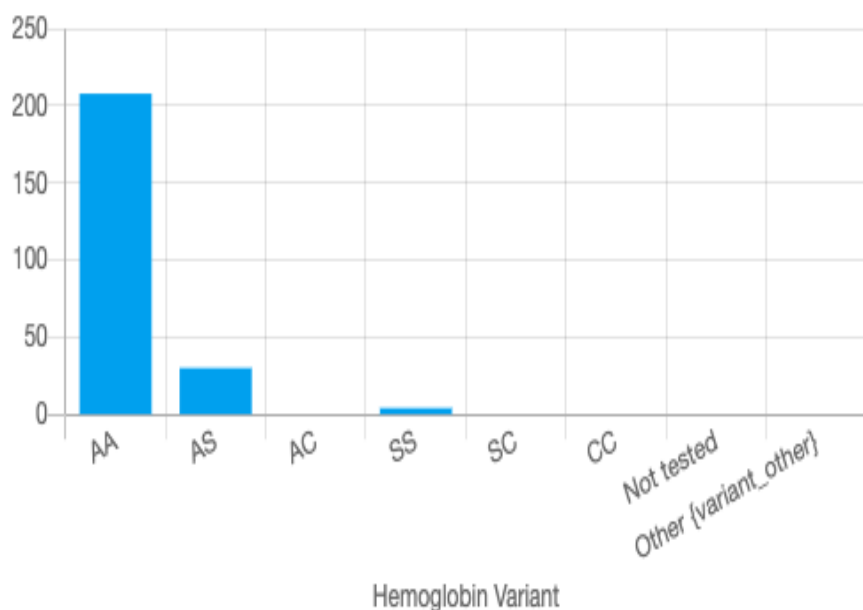
The screening identified a substantial burden of abnormal Hemoglobin variants:

- Approximately **86%** had **normal Hemoglobin (HbAA)**.
- Approximately **12%** were identified with **sickle cell trait (HbAS)**.
- Approximately **2%** were identified with **sickle cell disease (HbSS or other disease variants)**.

These findings indicate that **nearly one in seven participants had an abnormal Hemoglobin result**, emphasizing the value of routine screening among young adults.

RESULTS

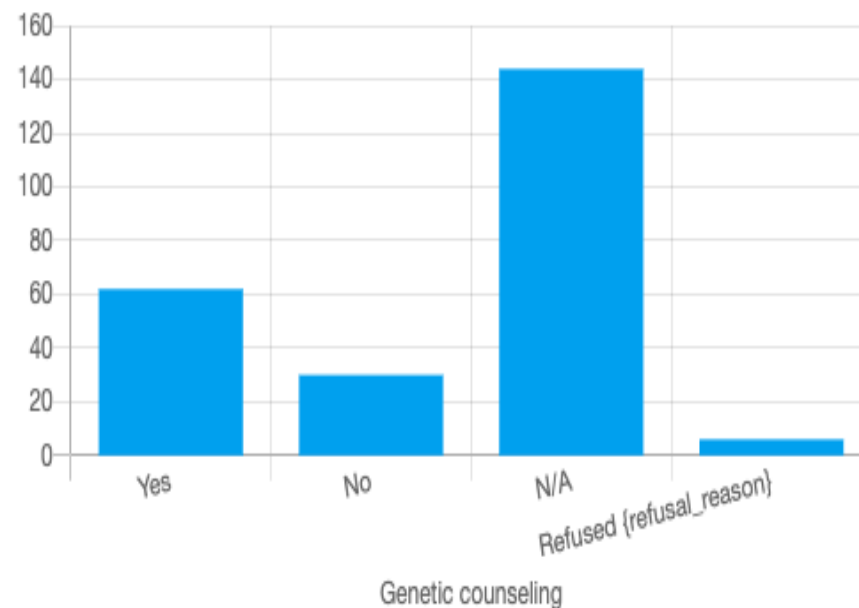
Hemoglobin Testing



Around 86% of the cohort had normal hemoglobin variant, **12%** tested for **sickle cell trait** (AS, no other variants were identified), and **nearly 2%** of test results were **sickle cell anemia**.

Potential Actions Taken

Genetic Counseling



[Reasons for refusal can be viewed here](#)

Refusal report

Age (years)	Hemoglobin Variant	Other variant	Genetic counseling	Refusal reason	Gender:	Participant Occupation
22	AS		Refused	knows family members have SCD	Female	Student
20	AA		Refused	understands results	Female	Student
22	AA		Refused	not needed	Male	Student
23	AA		Refused	not needed	Male	Student
43	AA		Refused	rushing back to the office	Female	Faculty
21	AA		Refused	studying community health and understands results	Female	Student

Post-Screening Actions

The initiative provided immediate follow-up support through:

- Genetic counseling
- Linkage to appropriate healthcare services
- Health education and awareness

Most participants indicated they understood the educational messages delivered, while those with abnormal screening results were offered additional counseling and referral.

Strategic Implications for JOOUST

The results demonstrate that JOOUST has an opportunity to become a regional and national leader in student-centered sickle cell prevention and awareness. The campaign identified both a measurable burden of sickle cell trait and disease and important gaps in insurance coverage and genotype awareness among students.

The findings support integrating SCD education into student wellness programmes and strengthening access to genetic counseling and follow-up services.

Recommendations

Immediate (Within 12 Months)

- Institutionalize annual SCD screening during student orientation and health weeks.
- Strengthen access to confidential genetic counseling.
- Ensure all students with abnormal screening results receive confirmatory testing and clinical follow-up.
- Increase awareness of genotype testing and reproductive health planning.

Medium-Term

- Integrate SCD education into university wellness and health promotion programmes.
- Collaborate with county health departments and referral hospitals to establish referral pathways.

- Promote enrolment in health insurance schemes to improve access to specialized care.

Long-Term

- Position JOOUST as a regional center of excellence for SCD awareness, screening, research, and community engagement.
- Expand partnerships with organizations such as AMPATH Kenya, the Ministry of Health, and patient advocacy groups to scale screening and education initiatives.

Conclusion

The JOOUST Sickle Cell Disease Screening and Education Initiative successfully demonstrated the feasibility and value of campus-based SCD screening. The identification of approximately **12% of participants with sickle cell trait** and **2% with sickle cell disease**, coupled with low insurance coverage and limited genotype awareness, underscores the need for sustained institutional investment in prevention, education, counseling, and linkage to care. By building on this initiative, JOOUST can strengthen student health services while contributing to Kenya's broader efforts to reduce the burden of sickle cell disease.

List of Contributors

- *Prof. George Ayodo: Jaramogi Oginga Odinga University Science and Technology (JOOUST) – Lead*
- *Dr. Samuel Mbunya, DBA: Innovative Hematology Inc., (IHI)/Dr. Mbunya Health and Leadership Consultancy – Lead*
- *Esther Bosibori Nyachieo: Jaramogi Oginga Odinga University Science and Technology (JOOUST)*
- *Eveline Moraa Masira: Jaramogi Oginga Odinga University Science and Technology (JOOUST)*
- *Winnie Kibwana: Jaramogi Oginga Odinga University Science and Technology (JOOUST)*
- *Linnet Nyaboke Mose: Jaramogi Oginga Odinga University Science and Technology (JOOUST)*
- *Kisha Hampton: Innovative Hematology Inc., (IHI)*
- *Dr. Isaac Jansen, PhD: Innovative Hematology Inc., (IHI)*
- *Erick Ayaye: Academic Model Providing Access to Healthcare (AMPATH Kenya)*
- *Phionah A. Obanda: Don Amolo Memorial Kids Ark (DAMKA)*
- *Veronica Bitta – Tumaini Sickle Cell Organization*
- *Rebecca Aluoch Jabuya – Siaya Sickle Cell Advocacy Network*
- *Dan Ogola – Matibabu Foundation*
- *Dr. Moses Elvis Oburah: Journey's by Grace Global Mission Kenya*
- *Alex Gruzdev: Silver Lake Research*

Photo Entry



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